Solution of Lagrange's linear PDE.

The general solution of the linear PDE.

$$Pp + Qq = R \quad \text{is} \quad \phi(u,v) = 0$$

where ϕ is an arbitrary function and $u(x,y,z) = c_1$ and $v(x,y,z) = c_2$ form a solution of the equation

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

which is called subsidiary equation.

Ex: Find the general integral of

$$xp + yq = z$$

Step 1. If we compare $Pp + Qq = R$ and $xp + yq = z$

we get $x = P \quad Q = y \quad R = z$

Step 2 Form the auxiliary simultaneous equations

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \quad (\Rightarrow) \quad \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$$

solve these auxiliary simultaneous equations

$$\begin{cases} \frac{dx}{x} = \frac{dy}{y} \\ \frac{dy}{y} = \frac{dz}{z} \end{cases}$$

$$\int \frac{dx}{x} = \int \frac{dy}{y} \Rightarrow \ln|x| = \ln|y| + C_1$$

$$\ln|x| - \ln|y| = C_1$$

Exponentiating both sides give

$$\left|\frac{x}{y}\right| = e^{C_1}$$ ↳ is a constant

so $$\frac{x}{y} = C_1$$

$$\int \frac{dy}{y} = \int \frac{dz}{z} \Rightarrow \ln|y| = \ln|z| + C_2$$

$$\ln|y| - \ln|z| = C_2$$

$$\ln\left|\frac{y}{z}\right| = C_2$$

Exponentiating both sides gives $$\left|\frac{y}{z}\right| = e^{C_2}$$ ↳ constant

so $$\frac{y}{z} = C_2$$

two independent solutions

$$u(x,y) = \frac{x}{y}$$
$$= c_1$$

and $v(y,z) = \frac{y}{z}$

$$= c_2$$

then the solution is written as

$$\phi(u,v) = \phi\left(\frac{x}{y}, \frac{y}{z}\right)$$

$$= 0$$

Method of Multipliers

This method help us to solve subsidiary equations

Starting from $\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$

let us introduce a common parameter ds

we may write

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = ds}$$

Thus

$$dx = P\,ds \qquad P = \frac{dx}{ds}$$

$$dy = Q\,ds \qquad Q = \frac{dy}{ds}$$

$$dz = R\,ds \qquad R = \frac{dz}{ds}$$

multiply the first equation by l, the second by m and the third by n, and add them

$$\boxed{l\,dx + m\,dy + n\,dz = (lP + mQ + nR)\,ds}$$

$$\therefore \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR} = ds$$

$$\Rightarrow \boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l\,dx + m\,dy + n\,dz}{lP + mQ + nR}.}$$

the same reasoning gives

$$\boxed{\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} = \frac{l'\,dx + m'\,dy + n'\,dz}{l'P + m'Q + n'R}.}$$

the multipliers l, m, n, l', m', n' may be constants or suitably chosen functions of x, y, z.

Choosing l, m, n such that

$$lP + mQ + nR = 0$$, we have

$$ldx + mdy + ndz = 0 \quad \text{— (*)}$$

~~If~~ * If $ldx + mdy + ndz$ is a perfect differential of some function, say $U(x, y, z)$ then

$$dU = 0$$ by (*)

~~by(*)~~ Hence integrating (*) we get

$$U = c_1$$

Similarly, the other set of multipliers l', m', n' can be found out so that

$$l'P + m'Q + n'R = 0$$

Hence $l'dx + m'dy + n'dz = 0$

If this expression is integrable, it gives the second integral $V = c_2$

therefore the general soln is

$$\boxed{\phi(U, V) = 0 \text{ or } U = f(V)}$$

Here, the set of multipliers l, m, n and l', m', n' are called Lagrangian multipliers.

Ex. Find the general integral solution of $px+qy=z$

To solve the equation

$$Pp+Qq=R$$

a) form the auxiliary simultaneous equations

$$\frac{dx}{P}=\frac{dy}{Q}=\frac{dz}{R}$$

b) Solve these auxiliary simultaneous equations giving two independent solutions

$$u=c_1 \text{ and } v=c_2$$

c) then write down the solution as

$$\phi(u,v)=0 \quad \text{or} \quad u=f(v) \text{ or } v=f(u)$$

where the f_1 is arbitrary.

Ex. 1) Find the general integral solution of

$$xp + yq = z$$

Sln comparing the equation with

$$Pp + Qq = R$$

we get $P = x$

$Q = y$

$R = z$

The subsidiary equations are

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{z}$$

from $\frac{dx}{x} = \frac{dy}{y}$ we get

$$\ln x = \ln y + \ln a$$

$$\frac{x}{y} = a$$

Similarly from $\frac{dy}{y} = \frac{dz}{z}$

$$\ln y = \ln z + \ln b$$

$$\frac{y}{z} = b$$

Hence the general integral solution is

$$\phi\left(\frac{x}{y}, \frac{y}{z}\right) = 0$$

EX2 Find the general solution of

$$x(z^2-y^2)p + y(x^2-z^2)q = z(y^2-x^2)$$

Sln compare this equation with

$$Pp + Qq = R.$$

$$P = x(z^2-y^2) \qquad Q = y(x^2-z^2)$$

$$R = z(y^2-x^2)$$

As $lP + mQ + nR = 0$

$\Leftrightarrow$ $lx(z^2-y^2) + my(x^2-z^2) + nz(y^2-x^2) = 0$

The subsidiary equations are

$$\frac{dx}{x(z^2-y^2)} = \frac{dy}{y(x^2-z^2)} = \frac{dz}{z(y^2-x^2)} \qquad (*)$$

Taking the two sets of multipliers as x, y, z and $\frac{1}{x}, \frac{1}{y}, \frac{1}{z}$ each of ratio in $(*)$, we get

$$\frac{x\,dx+y\,dy+z\,dz}{x^2(z^2-y^2)+y^2(x^2-y^2)+z^2(y^2-x^2)} = \frac{\frac{1}{x}dx+\frac{1}{y}dy+\frac{1}{z}dz}{(z^2-y^2)+(x^2-y^2)+(y^2-x^2)}$$

$$\frac{x\,dx+y\,dy+z\,dz}{0} = \frac{\frac{1}{x}dx+\frac{1}{y}dy+\frac{1}{z}dz}{0}$$

Thus $x\,dx+y\,dy+z\,dz=0$

and $\frac{1}{x}dx+\frac{1}{y}dy+\frac{1}{z}dz=0$

by integrating, we get

$$\int x\,dx+\int y\,dy+\int z\,dx = \frac{x^2+y^2+z^2}{2}$$

$$= C_1$$

$$\Rightarrow x^2+y^2+z^2 = 2c_1$$

and $\int \frac{1}{x}dx+\int \frac{1}{y}dy+\int \frac{1}{z}dz = \ln x+\ln y+\ln z = c_2$

$$\Rightarrow xyz = e^{c_2}$$

$$= C_3$$

the general solution is $\phi(x^2+y^2+z^2, xyz)=0$

General form of second order PDEs

$$A U_{xx} + 2B U_{xy} + C U_{yy} + D U_x + E U_y + F U = f.$$

If $f = 0$ we have homogeneous

$f \neq 0$ we have non-homogeneous

A, B, C, D, E and F are coefficients and are function of x and y.

f also is a fn of x and y.

Ex: 1) $(x+2y)U_x + x^2 U_y = \sin(x^2+y^2)$: non-homogeneous partial differential equation of first order

2) $(x+2y)U_x + x^2 U_y = 0$: homogeneous PDE of first order

3) $x U_{xx} + y U_{xy} + U_{yy} = \sin x$: non-homogeneous linear PDE of second order

Classification of the second order PDEs

Having $Au_{xx} + 2Bu_{xy} + Cu_{yy} + Du_x + Eu_y + Fu = 0$

we may have its algebraic quadratic equation, where we replace

u_x by α

u_y by β

u_{xx} by α^2

u_{xy} by $\alpha\beta$

and

u_{yy} by β^2

$P(\alpha,\beta) = A\alpha^2 + 2B\alpha\beta + C\beta^2 + D\alpha + E\beta + F = 0$

this quadratic equation represents a hyperbola, parabola, or ellipse according to its discriminant $B^2 - AC$.

If $B^2 - AC > 0$ we have hyperbolic

$B^2 - AC = 0$ " parabolic

$B^2 - AC < 0$ " Elliptic

Homogeneous constant coefficient PDEs

Define $D_x = \frac{\partial}{\partial x}$

$D_y = \frac{\partial}{\partial y}$

we are looking for solving equation of the type

$$\frac{\partial^2 u}{\partial x^2} + k_1 \frac{\partial^2 u}{\partial x \partial y} + k_2 \frac{\partial^2 u}{\partial y^2} = 0 \quad (*)$$

where k_1 and k_2 are constants.

(*) can be written as

$$D_x^2 + k_1 D_x D_y + k_2 D_y^2)u = 0$$

or $F(D_x, D_y)u = 0$

The auxiliary equation is obtained by replacing

$D_x \longrightarrow m$

$D_y \longrightarrow 1$

Thus $\boxed{m^2 + k_1 m + k_2 = 0}$ (**)

Distinct Roots

once we solve (**) we obtain two values

$m = m_1$

$m = m_2$

These values allow us to factorise the original PDE.

using the two roots, we rewrite equation (**) as

$$(D_x - m_1 y)(D_x - m_2 D_y)u = 0 \quad (***)$$

this is the factorised form of the second-order PDE.

Let us solve the first order PDE.

$$1)\left(D_x - m_2 D_y\right)u = 0$$

$$\frac{\partial u}{\partial x} - m_2 \frac{\partial u}{\partial y} = 0$$

$$P = \frac{\partial u}{\partial x}$$

$$q = \frac{\partial u}{\partial y}$$

Thus the equation becomes

$$P - m_2 q = 0$$

This is a first order linear PDE.

Recall that a first order PDE of the form

$$Pp + Qq = R$$

has the characteristic auxiliary system

$$\frac{dx}{P} = \frac{dy}{Q} = \frac{dy}{R}$$

for our equation

$$p - m_2 q = 0$$

$$P = 1 \quad Q = -m_2 \quad , \quad R = 0$$

substituting these values gives

$$\frac{dx}{1} = \frac{dy}{-m_2} = \frac{dy}{0}$$

Solve $\frac{dx}{1} = \frac{dy}{-m_2} \iff -m_2 dx = dy$

$$\iff dy + m_2 dx = 0$$

$$\Rightarrow \int dy + m_2 \int dx = 0$$

$$\Rightarrow \boxed{y + m_2 x = c}$$

this means that along each characteristic curve, the quantity

$y + m_2 x$ remains constant.

$y = -m_2 x + c$ this represents straight lines

now solve.

$$\frac{dx}{1} = \frac{dy}{0}$$

$$\Rightarrow du = 0$$

$$\int du = \int 0$$

$$\boxed{u = c_1}$$

u remains constant along each characteristic

The value of u need not be the same on every characteristic curve

Each characteristic is identified by the constant

$$\boxed{c = y + m_2 x}$$

consequently, C_1 can depend on C

$$\Rightarrow C_1 = \phi(C)$$

since $C = y + m_2 x$

we obtain

$$\boxed{u_1(x,y) = \phi(y + m_2 x)}$$

This is the general solution of the first order PDE.

similarly solve

$$(D_x - m_1 D_y)\, u = 0$$

$$\Rightarrow \frac{\partial u}{\partial x} - m_1 \frac{\partial u}{\partial y} = 0$$

$$\Leftrightarrow P - m_1 q = 0$$

$$P = 1 \quad Q = -m_1 \quad R = 0$$

Thus, the auxiliary system becomes

$$\frac{dx}{1} = \frac{dy}{-m_1} = \frac{dy}{0}$$

(1) $\frac{dx}{1} = \frac{dy}{-m_1}$

$\Rightarrow dy + m_1 dx = 0$

$\Rightarrow \int dy + m_1 \int dx = 0$

$\Rightarrow \boxed{y + m_1 x = c_2}$

(2) $du = 0$

$u = c_3$ c_3 depend on c_2

as before

$c_3 = \psi(c_2)$

Substituting c_2, we obtain

$\boxed{u_2(x,y) = \psi(y + m_1 x)}$

the general soln of second first-order pde

Thus the solution is

$u_2 u_1 + u_2$

$$\Rightarrow \boxed{u(x,y) = \phi(y + m_2 x) + \psi(y + m_1 x).}$$

ϕ and ψ are arbitrary twice continuously differentiable.